

Action History: “I Have Seen This Before” Is Not the Same as “I Did This Before”

Human-readable companion to Action-Specific History in Present Choice and Initiation: Separating Provenance Discriminability from Repeated-Action History

Human-readable companion to “Action-Specific History in Present Choice and Initiation: Separating Provenance Discriminability from Repeated-Action History.” This companion preserves the scientific claims, limits, and contraction rules of the source manuscript while foregrounding the ordinary human question that motivated it. It is not a separate research contribution.

Abstract

The present can look the same while its relation to a person’s past actions differs. A familiar object may be easy to recognize but hard to place: did I choose it, reject it, or only see it? A repeated routine may present no source-memory problem at all, yet the same current action can sit inside a shallow or deep history of doing that action in this context. PhoneWalletKeys separates these two forms of action-specific history instead of treating “history” as one hidden variable. Test A manipulates how discriminable self-chosen options are from equally familiar seen-but-not-chosen options and asks whether later option weighting changes after source discrimination and generic recognition are independently validated. Test B manipulates context-specific accumulated history of performing the same action, crosses that history with anticipated future recurrence, and measures bounded self-timed onset alongside a timing-matched externally triggered comparison. Repetition is not assumed to inhibit action: facilitation, inhibition, and null outcomes are all legitimate. The two tests may have entirely different lower-level mechanisms. Their possible unity is an experimental discipline for holding the present action problem sufficiently fixed while manipulating a declared relation to one’s own prior action.

1. Two ordinary problems that both involve “what I did before”

One familiar problem is provenance. You recognize every item in front of you, but that does not tell you which one you already selected. “I have seen this” and “I chose this” are different pieces of information. A self-ordered task can therefore become difficult even when generic familiarity is excellent.

A second problem is almost the reverse. There may be no uncertainty at all about the past action. You have performed the same routine repeatedly in the same context. The current action itself is familiar and obvious; the open question is whether the accumulated history associated with that context changes how readily you enter the action now.

The paper calls both action-specific historical relations because both connect a present action problem to information about the person’s own prior actions. It does not claim that they are one mechanism, one memory store, or one latent quantity.

2. “History matters” is already established and is not the contribution

Past behavior influences future behavior through many established routes: source monitoring, episodic retrieval, habit, context-response learning, value, recent motor history, boredom, variety seeking, construal, expectation, and more. PhoneWalletKeys starts by conceding that broad territory.

The residual question is experimental rather than metaphysical: can two narrower properties of self-action history be manipulated separately while the immediate action problem is kept sufficiently stable, and can each manipulation make an independently testable contribution to current choice or initiation?

3. Test A: separate familiarity from self-action provenance

Test A uses three item classes. C items are familiar and were previously self-chosen or acted on. S items are equally familiar but were seen or encountered without being self-chosen. N items are genuinely new. The critical comparison is C versus S: both are old, but only one class has the relevant self-action source relation.

Class	Current familiarity	Prior self-action relation	Why it is needed
C	Old / familiar	Previously self-chosen or acted on	Tests weighting of options with self-action provenance.
S	Old / familiar	Previously seen but not self-chosen	Separates source identity from generic oldness.
N	New	No prior encounter	Provides a new-option comparison; it is not automatically “exploration.”

The manipulation occurs at encoding. High- versus low-source-diagnostics conditions change how distinguishable the C and S encounters are without intentionally changing their generic familiarity, current value, exposure dose, or other present-choice properties. Later, source discrimination and ordinary recognition are checked separately before source effects are interpreted.

4. What Test A can establish - and where it fails

The minimal Test A prediction is that a valid source-diagnostics manipulation changes later weighting between C and S. A stronger subordinate question asks whether low provenance discriminability also shifts relative weighting toward N after generic recognition and value are sufficiently controlled.

That stronger novelty-related effect is not required for the branch to survive. If only C-versus-S weighting changes, a repeat-choice, choice-supportive reconstruction, confidence, or source-memory account may be sufficient. The branch simply contracts to the narrower provenance-to-choice relation.

- Test A fails if source discrimination changes but present option weighting does not.
- The implementation is invalid if source diagnostics cannot be separated from material changes in generic item recognition or current value.
- Novel-option selection is not labeled directed exploration unless information seeking is independently operationalized.
- A final choice cannot identify the unique internal route; source reconstruction, confidence, retrieval fluency, demand, and value remain alternative paths.

5. Test B: hold the present action fixed and change its context-specific repeated history

Test B asks a different question. The current action is kept the same while its learned relation to context and, where feasible, the depth of previous same-action repetition are varied. The critical historical variable is not simply “have you done this before?” but how much repeated performance of this same action is associated with the context in which the present trial occurs.

Past repetition is crossed with anticipated future recurrence. This prevents “I have already done this many times” from being silently confounded with “I expect I will have to keep doing this many more times.” Those may feel similar in ordinary life, but they are different experimental relations.

6. Why self-timed onset needs an externally triggered comparison

The primary outcome is bounded self-timed onset: within a defined response window, when does the participant initiate the action without an external go cue dictating the exact moment? The design also includes a temporally yoked externally triggered comparison. That branch helps determine whether a history effect is specific to self-timed entry or simply changes execution speed, preparation, motor fluency, or generic response latency.

past same-action history x expected future recurrence -> self-timed onset, compared with matched external triggering

The comparison is an identification aid, not proof of a hidden initiation module. Selectivity at one endpoint still leaves several ordinary mechanisms available.

7. Test B deliberately does not predict that repetition must make action harder

The motivating mundane observation may sound like accumulated repetition should make a task harder to re-enter. The manuscript refuses to build that direction into the theory. Repetition often facilitates action through habit, context-response learning, retrieval fluency, or recent motor history.

Observed result	What it would support	How the interpretation contracts
Longer self-timed onset with deeper history	The motivating resistance/inhibition direction under the tested controls.	Still requires discrimination from boredom, value, expectation, monotony, obligation, and related conventional routes.
Shorter self-timed onset with deeper history	Sensitivity to repeated history in a facilitative direction.	The motivating resistance interpretation contracts toward habit, fluency, or motor-history explanations.
No history effect	No support for a repeated-history onset relation under the tested conditions.	The Test B claim is killed for that implementation.
Only future recurrence matters	Prospective expectation/obligation predicts onset, not accumulated history.	The historical claim contracts accordingly.

8. Why Test A and Test B are allowed to remain separate

The paper does not force a common lower-level mechanism. Test A may be explained by source monitoring, episodic reconstruction, confidence, or value-linked memory. Test B may be explained by context-response learning, habit, planning, boredom, construal, expectancy, or recent motor history. Success in one branch does not validate the other.

Their proposed unity is higher-level and methodological. In each branch, the experimenter declares a relation between the present problem and the person's prior actions, manipulates that relation while trying to keep the present problem stable, independently validates the historical manipulation, measures a current consequence, and keeps explicit failure conditions rather than inferring an invisible "history force."

9. The composition itself can fail

- If the only common conclusion is "past behavior affects future behavior," the paper-level composition adds nothing; that proposition is already established.
- If Test A and Test B require unrelated definitions, controls, failure logic, and no useful common distinction between present action problem and historical relation, they should be treated as separate conventional effects.
- If direct prior art already assembles substantially the same two functions at the same level, the paper-level contribution is subsumed.
- If experiments show that the common historical organization adds no prospective explanatory or predictive value beyond task-specific models, it should contract.

This is why the paper avoids a single latent history variable. The experimental distinction has to earn its keep before any stronger theoretical unity is preserved.

10. The point of PhoneWalletKeys

A current action can be related to the past in more than one way. Familiarity can be high while self-action provenance is uncertain. Provenance can be clear while accumulated repetition associated with the current context changes how readily the same action is entered. Those are testable relations even if familiar cognitive mechanisms fully explain them.

PhoneWalletKeys therefore does not claim a new memory system or a universal cost of repetition. It asks the experimenter to name which historical relation is being manipulated, validate it independently of the endpoint, and let each branch die if the predicted present-action consequence is absent. The broad claim "history matters" is intentionally too weak to count as success.

Selected References

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